

MAT 3150 — Homework 4

Practice: Same Techniques, Different Problems

Each section practices the technique behind one homework problem. The **worked example** is fully solved; the **practice** problems are for you to try (hints only).

1 Telescoping and step-by-step bounds (for HW Problem 1)

Tools.

- Telescoping: $x_{n+1} - x_N = (x_{n+1} - x_n) + (x_n - x_{n-1}) + \cdots + (x_{N+1} - x_N)$.
- Adding inequalities: if $a_1 < b_1$, $a_2 < b_2$, $\dots$, $a_m < b_m$, then $a_1 + \cdots + a_m < b_1 + \cdots + b_m$.
- Multiplying by $c > 0$ keeps the direction of $<$.

Worked example

Prove: if $1 < x_{k+1} - x_k < 3$ for every $k \geq 1$, then

$$n < x_{n+1} - x_1 < 3n \quad \text{for every } n \geq 1.$$

Solution. Write the one-step inequality for $k = 1, 2, \dots, n$:

$$\begin{aligned} 1 &< x_2 - x_1 < 3 \\ 1 &< x_3 - x_2 < 3 \\ &\vdots \\ 1 &< x_{n+1} - x_n < 3. \end{aligned}$$

There are n lines. Add them. The left sides add to $n \cdot 1 = n$, the right sides to $n \cdot 3 = 3n$, and the middle telescopes:

$$(x_2 - x_1) + (x_3 - x_2) + \cdots + (x_{n+1} - x_n) = x_{n+1} - x_1,$$

because every $x_2, \dots, x_n$ appears once with $+$ and once with $-$. Hence $n < x_{n+1} - x_1 < 3n$. ✓

Practice 1A

Compute $(y_5 - y_4) + (y_4 - y_3) + (y_3 - y_2)$. Then compute $(y_{n+1} - y_n) + (y_n - y_{n-1}) + \cdots + (y_{N+1} - y_N)$ for general $N < n$.

Practice 1B

Let (y_n) be increasing. **Prove:** if $2(y_{k+1} - y_k) < x_{k+1} - x_k$ for every $k \geq 5$, then

$$2(y_{n+1} - y_5) < x_{n+1} - x_5 \quad \text{for every } n \geq 5.$$

Hint: same as the worked example, but now the left sides are not all equal. Factor out the 2 before you add, then telescope inside the bracket.

Practice 1C

Use telescoping to show $\sum_{k=1}^n \frac{1}{k(k+1)} = 1 - \frac{1}{n+1}$.

Hint: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$.

Setup. First check the hint by combining over a common denominator:

$$\frac{1}{k} - \frac{1}{k+1} = \frac{(k+1) - k}{k(k+1)} = \frac{1}{k(k+1)}.$$

Now replace every term of the sum by its split form and write the terms out one per k :

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k(k+1)} &= \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= \underbrace{\left(\frac{1}{1} - \frac{1}{2} \right)}_{k=1} + \underbrace{\left(\frac{1}{2} - \frac{1}{3} \right)}_{k=2} + \underbrace{\left(\frac{1}{3} - \frac{1}{4} \right)}_{k=3} + \cdots + \underbrace{\left(\frac{1}{n-1} - \frac{1}{n} \right)}_{k=n-1} + \underbrace{\left(\frac{1}{n} - \frac{1}{n+1} \right)}_{k=n}. \end{aligned}$$

Cancel. Each of $\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}$ appears exactly twice: once with a minus sign (as the second part of one bracket) and once with a plus sign (as the first part of the next bracket). For example, $-\frac{1}{2}$ from $k=1$ cancels $+\frac{1}{2}$ from $k=2$, and $-\frac{1}{n}$ from $k=n-1$ cancels $+\frac{1}{n}$ from $k=n$. The only terms that never get a partner are the very first, $+\frac{1}{1}$, and the very last, $-\frac{1}{n+1}$.

Finish. Therefore

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{1}{1} - \frac{1}{n+1} = 1 - \frac{1}{n+1}. \quad \checkmark$$

Check with $n=2$. Directly: $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$. Formula: $1 - \frac{1}{3} = \frac{2}{3}$. They agree.

Connection to HW Problem 1. The same collapse happens in $(x_{n+1} - x_n) + (x_n - x_{n-1}) + \cdots + (x_{N+1} - x_N)$: only the first term of the first bracket and the last term of the last bracket survive.

2 Squeeze, divide, and kill the leftovers (for HW Problem 2)

Tools.

- If C is a fixed number and $y_n \rightarrow \infty$, then $C/y_n \rightarrow 0$.
- If you need two conditions “for $n \geq N_1$ ” and “for $n \geq N_2$ ”, take $n \geq \max(N_1, N_2)$.

Worked example

Prove: if $2n - 7 < x_n < 2n + 4$ for all $n \geq 10$, then $\lim_{n \rightarrow \infty} \frac{x_n}{n} = 2$.

Solution. For $n \geq 10$ we have $n > 0$, so dividing by n keeps the inequalities:

$$2 - \frac{7}{n} < \frac{x_n}{n} < 2 + \frac{4}{n}.$$

Subtract 2:

$$-\frac{7}{n} < \frac{x_n}{n} - 2 < \frac{4}{n}, \quad \text{so} \quad \left| \frac{x_n}{n} - 2 \right| < \frac{7}{n}.$$

The leftover terms $\frac{7}{n}$ and $\frac{4}{n}$ come from *fixed* constants divided by something going to ∞ , so they can be made small.

Let $\varepsilon > 0$. Choose $N_2 \in \mathbb{N}$ with $N_2 > \frac{7}{\varepsilon}$ (Archimedean property), and let $M = \max(10, N_2)$. For $n \geq M$, both the squeeze (needs $n \geq 10$) and $\frac{7}{n} < \varepsilon$ (needs $n \geq N_2$) hold, so

$$\left| \frac{x_n}{n} - 2 \right| < \frac{7}{n} \leq \frac{7}{M} < \varepsilon.$$

Hence $x_n/n \rightarrow 2$. ✓

Practice 2A

Prove from the definition: if $C \in \mathbb{R}$ is fixed and $y_n \rightarrow \infty$, then $C/y_n \rightarrow 0$.

Hint: you want $|C|/y_n < \varepsilon$, i.e. $y_n > |C|/\varepsilon$. Which definition gives you that? (Handle $C = 0$ separately, or note it's trivial.)

Practice 2B

Prove: if $n^2 - 5n < x_n < n^2 + 3$ for all $n \geq 1$, then $\lim_{n \rightarrow \infty} \frac{x_n}{n^2} = 1$.

Hint: divide by n^2 , subtract 1, and look at which leftover is bigger in absolute value.

Practice 2C

Let $y_n \rightarrow \infty$, $y_n > 0$, and fix numbers a, b . **Prove:** if $(3 - \varepsilon)y_n + a < x_n < (3 + \varepsilon)y_n + b$ for all large n , then for all large n ,

$$3 - 2\varepsilon < \frac{x_n}{y_n} < 3 + 2\varepsilon.$$

Hint: divide by y_n ; then use Practice 2A to make a/y_n and b/y_n smaller than ε in absolute value.

3 Using a theorem by choosing x_n and y_n (for HW Problem 3)

Tool (HW Problem 2, Stolz–Cesàro). If (y_n) is increasing, $y_n \rightarrow \infty$, and $\frac{x_{n+1} - x_n}{y_{n+1} - y_n} \rightarrow L$, then $\frac{x_n}{y_n} \rightarrow L$.

Worked example

Prove: $\lim_{n \rightarrow \infty} \frac{1 + 2 + \cdots + n}{n^2} = \frac{1}{2}$.

Solution. Let $x_n = 1 + 2 + \cdots + n$ and $y_n = n^2$.

- (y_n) is increasing since $(n+1)^2 > n^2$, and $y_n = n^2 \rightarrow \infty$.
- $x_{n+1} - x_n = n + 1$ (only the last term survives), and $y_{n+1} - y_n = (n+1)^2 - n^2 = 2n + 1$.
- So $\frac{x_{n+1} - x_n}{y_{n+1} - y_n} = \frac{n+1}{2n+1} \rightarrow \frac{1}{2}$ (divide top and bottom by n).

By Stolz–Cesàro, $\frac{x_n}{y_n} \rightarrow \frac{1}{2}$. ✓

Practice 3A

Prove: $\lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + \cdots + n^2}{n^3} = \frac{1}{3}$.

Hint: $(n+1)^3 - n^3 = 3n^2 + 3n + 1$.

Practice 3B

Prove: if $b_n \rightarrow 5$, then $\lim_{n \rightarrow \infty} \frac{b_1 + b_2 + \cdots + b_n}{n^2} = 0$.

Hint: what are $x_{n+1} - x_n$ and $y_{n+1} - y_n$ now? Does a convergent sequence divided by something going to ∞ go to 0?

4 Monotone and bounded recursive sequences (for HW Problem 4)

Tools.

- Monotone Convergence Theorem: a monotone, bounded sequence converges.
- Induction: prove a base case, then “true for $n \Rightarrow$ true for $n + 1$ ”.
- If $a_n \rightarrow a$ and $a_{n+1} = f(a_n)$ with f continuous at a , then $a = f(a)$ (since $a_{n+1} \rightarrow a$ too).

Worked example

Let $a_1 = 2$ and $a_{n+1} = 4 - \frac{3}{a_n}$. Show (a_n) converges and find the limit.

Scratch work. $a_2 = 4 - \frac{3}{2} = 2.5$, $a_3 = 4 - \frac{3}{2.5} = 2.8$, $a_4 \approx 2.93$. Guess: increasing, and stays between 1 and 3.

Bounded: $1 < a_n < 3$ for all n . Base case: $a_1 = 2$. Inductive step: assume $1 < a_n < 3$. Then $\frac{1}{3} < \frac{1}{a_n} < 1$, so $1 < \frac{3}{a_n} < 3$, so

$$4 - 3 < 4 - \frac{3}{a_n} < 4 - 1, \quad \text{i.e.} \quad 1 < a_{n+1} < 3.$$

Increasing. Compute directly:

$$a_{n+1} - a_n = 4 - \frac{3}{a_n} - a_n = \frac{4a_n - 3 - a_n^2}{a_n} = -\frac{(a_n - 1)(a_n - 3)}{a_n}.$$

Since $1 < a_n < 3$: $a_n - 1 > 0$, $a_n - 3 < 0$, $a_n > 0$, so the fraction is negative and $a_{n+1} - a_n > 0$.

Converges. Increasing and bounded, so by the Monotone Convergence Theorem $a_n \rightarrow a$ for some a . Since $a_n \geq a_1 = 2$, we get $a \geq 2$; in particular $a \neq 0$.

The limit. Let $n \rightarrow \infty$ in $a_{n+1} = 4 - \frac{3}{a_n}$:

$$a = 4 - \frac{3}{a} \implies a^2 - 4a + 3 = 0 \implies a = 1 \text{ or } a = 3.$$

Since $a \geq 2$, $a = 3$. ✓

Practice 4A

Same recursion, different start: $a_1 = 5$, $a_{n+1} = 4 - \frac{3}{a_n}$. Show (a_n) converges and find the limit.

Hint: compute a_2, a_3 first. Is it still increasing? What bound do you expect now?

Practice 4B

Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. Show (a_n) converges and find the limit.

Hint: guess the bound $a_n < 2$. For monotone, compare a_{n+1}^2 and a_n^2 , or use induction: if $a_n < a_{n+1}$, is $a_{n+1} < a_{n+2}$?